

Convertibility Is Nearly Free: A Capable, Fully Tensor-Decomposable Vision-Language-Action Model

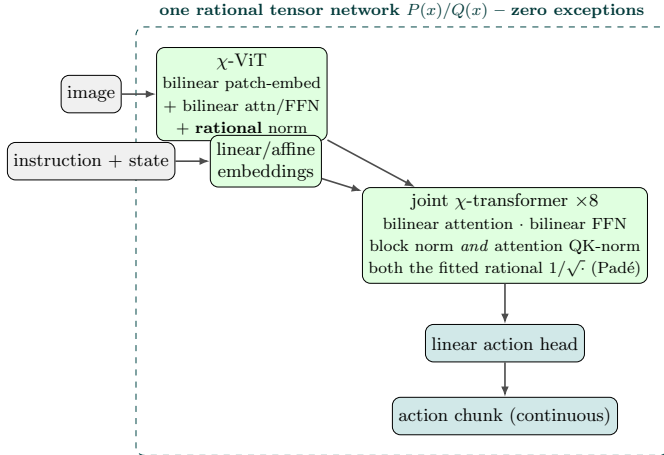
Convertibility is nearly free for language models, but not automatically for multimodal ones. We recover it: multimodal action decoding and rational per-instance normalization within the (rational) tensor-network class, with weight-based interpretability for free

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Abstract

A line of work replaces every nonlinearity in a network with bilinear/polynomial operations, so the whole learned map is a single tensor network that can be decomposed, and interpreted, directly from its weights. The unspoken price is *capability*: the operations a capable policy needs but a polynomial cannot express, multimodal action decoding, per-instance normalization, seem to force a softmax, a sampler, or an input-dependent division that leaves the tensor-network class. This tax was already shown to be small for *language* models, whose sole output nonlinearity is a categorical softmax that lives outside the learned map as a decode, but it does *not* transfer automatically to multimodal policies, whose output is a continuous, jointly multimodal action distribution rather than a categorical next-token pick. We show the tax is nonetheless largely recoverable, and build χ -VLA, a Vision-Language-Action policy (pixels + instruction $\rightarrow$ robot actions) whose learned blocks admit a fully tensor-decomposable polynomial or rational configuration. **(1) Multimodal control without leaving the tensor network.** If the learned action *distribution* is kept in-graph and polynomial and only the *decode* is moved outside (a routing sign, exactly the status of OpenVLA’s external argmax), a genuinely multimodal action head can be built *only* from CP cores, addressing the fact that a linear-mean head is an invalid “average of valid behaviours” wherever the demonstrations are multimodal ($\sim 40\%$ of states, which we measure). Such a head drives non-trivial end-to-end closed-loop control in the LIBERO-Object simulator. At our small, from-scratch scale we make *no* claim that the multimodal head outperforms the linear-mean head because their seed ranges overlap under matched evaluation. The contribution is representational, that a multimodal action distribution can be carried and controlled entirely in-graph. **(2) Per-instance normalization in the rational tensor-network class.** Mean-square normalization folds *exactly* in projective (numerator, denominator) coordinates into a ratio $P(x)/Q(x)$ of polynomial tensor networks, verified to 10^{-14} . A fixed rational approximation to inverse square root gives an RMS-like per-token operator within the same rational class. Together (1)–(2) recede both inherently-nonlinear costs behind one principle: *keep the learned object in-graph and polynomial/rational. Decode outside.* **(3) Interpretability comes for free, coherence is bought by topology.** The exact weight-derived orthogonalize–diagonalize–truncate (ODT) basis is causally faithful (matching PCA and integrated gradients, far exceeding random on deletion/insertion) and survives attention. Whether its atoms are *human-coherent* is set by topology, not convertibility (spatial-locality-preserving models reach $\sim 3.5\times$ random-baseline locality), confirming and, with interventions, matched capability, and a policy bond-placement principle, extending prior bilinear-interpretability work. Honest scope: the interpretability payoff is a causal, rankable, topology-tunable basis and a

locality *metric*, not yet a gallery of nameable features. Our strongest closed-loop checkpoint reaches $94.2\% \pm 1.2\%$ on eight of ten LIBERO-Object tasks, using a 600-step cap and per-token RMS normalization, its one remaining non-foldable operator. Replacing that operator with the rational per-instance norm and retraining from scratch makes every learned site a polynomial or rational tensor primitive by construction. This model reaches **94.5%** (189/200) on all ten LIBERO-Object tasks in one seed. No capability loss is visible in this point estimate, although the per-token and rational evaluations differ in suite coverage and seed count. The OpenVLA study reports $88.4\% \pm 0.8\%$ for OpenVLA and $92.5\% \pm 0.7\%$ for Diffusion Policy over all ten tasks with a 280-step cap [1]. Our evaluation uses a looser 600-step cap and fewer trials, so we make no state-of-the-art claim. We further verify decomposability on the trained checkpoint itself, not just architecturally: a mechanical audit confirms every module the policy actually invokes is polynomial or rational with zero exceptions, and an exact projective fold-and-verify on the real trained weights reproduces the deployed computation to the float32 precision floor (§4.5). A real causal intervention in the live simulator (swapping which co-present object the instruction names) gives an honest negative (switch-rate 15.6%, §6.5), which we attribute to real training data confounding scene identity with instruction, a limitation the synthetic-task intervention was built to avoid. The result establishes the narrower claim that rational tensor construction is compatible with competitive closed-loop control.



Model	Tasks	Success
χ -VLA, rational norm	10/10	94.5%
χ -VLA, per-token norm	8/10	$94.2\% \pm 1.2\%$
χ -VLA, conv vision	8/10	$90.8\% \pm 0.3\%$
Diffusion Policy [1]	10/10	$92.5\% \pm 0.7\%$
OpenVLA-7B [1]	10/10	$88.4\% \pm 0.8\%$

Closed-loop success, LIBERO-Object. Settings differ (step cap, trials/task, seeds – Table 3); not a matched state-of-the-art claim. χ -VLA is $\sim 20\text{M}$ parameters, trained from scratch, no pretraining.

Figure 1: **The current fully-tensor χ -VLA (left) and its best closed-loop results (right).** With a plain linear head, every learned computation – vision, joint attention backbone, normalization, and the action output itself – is one polynomial/rational tensor network; there is no out-of-graph decode step at all in this configuration (contrast the multimodal head, Fig. 4, whose sign-routing decode is the one exception).

1 Introduction

Interpreting a large policy is hard largely because the network is not compositional. Nonlinear operations such as the softmax in attention, the smooth activation in an MLP, and layer normalization prevent us from rewriting the model as a single algebraic formula, so we are forced to study its behaviour on data rather than its weights directly. A recent line of work argues that this is a design choice, not a necessity. If every learned operation is kept polynomial in the activations, built only from linear maps, element-wise multiplications, additions, fixed masks and permutations, learned constants, and fixed scalar or diagonal rescalings, then the whole model is a single multilinear object, a *tensor network*, that can be decomposed, ranked, and truncated using numerical linear algebra applied to the weights alone, no data required. The three subsections below state, in order, the question we ask, the prior results we build directly on, and what in this paper is new.

1.1 Goal

What does full tensor-decomposability actually cost a capable policy, and what does it buy? The standing worry is that forcing every operation to be polynomial trades away real capability for the promise of interpretability. This is an open question specifically for *vision-language-action* (VLA) policies, whose output is a continuous, jointly multimodal action distribution rather than a language model’s categorical next-token pick, and whose normalization sits inside a perception-heavy, degree-two-amplifying stack. We build χ -VLA, a VLA policy (pixels + instruction $\rightarrow$ robot actions) whose learned blocks admit a fully tensor-decomposable polynomial or rational configuration, and use it to answer the question directly: train it, close the loop in a real simulator, and interrogate its weights.

1.2 What this builds on

Convertibility, i.e. replacing every nonlinearity with a polynomial or bilinear primitive, was already shown to be nearly free for two narrower settings, and we assemble both into one policy.

Convertibility is nearly free for language models [2, 3]. A language model’s only output nonlinearity is the categorical softmax, and that softmax is a *decode*: linear logits go in, an arg max or a sample comes out, and the pick already lives outside the learned map. Making the rest of the network polynomial therefore costs almost nothing, because the one hard operation was never in the tensor network to begin with. Riggs [2] gives the two core primitives we reuse throughout, the bilinear (CP-factorized) feed-forward layer and softmax-free bilinear attention. Doods et al. [3] give foldable (frozen-scalar) normalization and Orthogonalize–Diagonalize–Truncate (ODT), the procedure for exactly, globally decomposing a tree-shaped tensor network, as a *compression* result.

The policy skeleton: OpenVLA [1]. We keep the modular OpenVLA pipeline (vision encoder(s) $\rightarrow$ projector $\rightarrow$ language-model backbone $\rightarrow$ action decoder) and replace each of its four learned stages with the corresponding published polynomial primitive:

- **Bilinear feed-forward layer** (Riggs [2]). Replace the gated MLP $D(\text{swish}(Lx) \otimes Rx)$ with the activation-free $D[(L\bar{x}) \otimes (R\bar{x})]$, where $\otimes$ is the element-wise (Hadamard) product and $\bar{x}$ is defined below. This is exactly a low-rank factorization (a CP, or canonical polyadic, factorization) of a third-order weight tensor.

- **Softmax-free bilinear attention** (Riggs [2]). Replace the softmax attention pattern with the element-wise product of two separate learned query-key score matrices.
- **Foldable normalization** (Dooms et al. [3]). Normalize during training by dividing by a running average magnitude (a single scalar), then after training freeze that scalar and multiply it into the next weight matrix, so the deployed model contains no division at all.
- **Linear action head.** Replace the discretized action tokens and language-model-style decoding with a direct linear map from learned action-query states to a continuous action vector.

We assemble these along the OpenVLA skeleton, keeping only genuinely non-learned interfaces outside the tensor-network boundary (tokenization, camera decoding, safety clipping, motor communication), so that χ -VLA’s learned map from pixels and token identities to a continuous action is, by construction, one tensor network.

1.3 What is novel here

Two of the four borrowed replacements above are *not* actually solved by the published primitives once the policy has to be capable, and the interpretability literature they enable has, so far, never been run on a real, working, closed-loop robot policy. Both gaps are what this paper adds.

Novelty 1: recovering the two operations that generically break convertibility. A linear action head trained by MSE can only emit a conditional *mean*, which is an invalid “average of valid behaviours” whenever demonstrations are multimodal ($\sim 40\%$ of LIBERO-Object states, which we measure), and the frozen-scalar norm only *approximates* true per-instance normalization. Neither failure is addressed by prior tensor-pure work, because neither arises for a language model. We show both are recoverable *within* the polynomial/rational tensor-network class, not by relaxing it: (i) a multimodal action head built entirely from CP cores, whose learned *distribution* stays in-graph and only a routing *sign* is decoded outside (the same status OpenVLA gives its argmax); and (ii) per-instance normalization carried in projective (numerator, denominator) coordinates, which folds *exactly* into a ratio $P(x)/Q(x)$ of two polynomial tensor networks, verified to 10^{-14} , with a fixed rational approximation to $1/\sqrt{\cdot}$ giving an RMS-like operator in the same class. One principle covers both: *keep the learned object in-graph and polynomial/rational, decode outside*.

Novelty 2: the interpretability payoff, measured on a real, competitive, closed-loop policy. Prior tensor-interpretability work stops at compression (Dooms et al. [3]) or asserts, without an intervention, that architecture affects eigen-feature coherence on shallow classifiers (Pearce et al. [2]; Déhérand [4]). We instead (i) separate *causal* from *coherent* as an explicit, intervention-tested axis with standard faithfulness baselines, (ii) establish topology, not convertibility, as the coherence lever at matched capability, with depth-scaling and cross-dataset de-risking, (iii) introduce a *bond-placement* principle for policies (a mechanism is decodable only at the bond downstream of the attention that constructs it), and (iv) run all of this on χ -VLA’s actual trained, closed-loop LIBERO-Object policy, not only on classifiers or a synthetic task.

Our contributions, established across substrates of increasing difficulty (image classifiers, a softmax-free vision transformer, the full tensor-decomposable policy, and its closed-loop control), are:

1. **Multimodal control without leaving the tensor network.** Keeping the learned action *distribution* in-graph and polynomial and moving only the *decode* outside, a genuinely multi-

modal action head can be built only from CP cores, since a linear-mean head is invalid wherever the demonstrations are multimodal ($\sim 40\%$ of states, measured). Such a head achieves non-trivial closed-loop success on the LIBERO-Object simulator. At our small, from-scratch scale the product and linear head seed ranges overlap, so we make no superiority claim. The contribution is that a multimodal action distribution is representable and controllable entirely in-graph.

2. **Per-instance normalization in the rational tensor-network class.** Carried in projective (numerator, denominator) coordinates, mean-square normalization folds *exactly* into $P(x)/Q(x)$ (both polynomial/CP), verified to 10^{-14} . A fixed rational inverse-square-root approximation gives an RMS-like operator without a frozen scalar. Contributions 1–2 recede both inherently nonlinear costs behind one principle: *learned object in-graph. Decode outside.*
3. **Interpretability is free. Coherence is bought by topology.** The exact weight-derived ODT basis is causally faithful by deletion/insertion (matching PCA and integrated gradients, far exceeding random) and survives attention. Whether its atoms are human-coherent is set by topology, not convertibility (spatial-locality-preserving topology reaches $\sim 3.5\times$ random-baseline locality, at no accuracy cost). We add interventions, matched-capability controls, and depth-scaling to prior bilinear interpretability, and are honest that this coherence is a *metric*, not yet a gallery of nameable features.
4. **For a policy, mechanisms live behind the attention.** The action-relevant quantity is constructed by attention and readable only at a post-attention bond, and decoding it there is not the same as controlling it.

Recovering novelty 1 (contributions 1–2), and closing the loop in simulation, is where the architecture stops being a mere substrate and becomes part of the contribution; novelty 2 (contributions 3–4) is the interpretability payoff that substrate is built to support.

2 Background: the three polynomial primitives

The homogeneous coordinate. Throughout, we write $\bar{x} = [1; x]$ for the input with a constant 1 prepended. This “homogeneous coordinate” lets a bilinear layer represent not only quadratic terms but also linear terms and constants (a bias becomes the weight on the constant coordinate). Without it, a stack of L multiplicative layers would be forced to produce only degree- 2^L terms. With it, the network can represent every degree up to 2^L , including the near-linear functions that most of a real task needs.

Bilinear feed-forward layer (a CP-factorized cubic tensor). With $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \otimes (R\bar{x})], \quad y_o = \sum_{i,j} \left(\underbrace{\sum_r D_{or} L_{ri} R_{rj}}_{T_{oij}} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The middle expression makes the tensor structure explicit: the layer is exactly a third-order weight tensor $T \in \mathbb{R}^{d \times (d+1) \times (d+1)}$ contracted twice with the input. The factored form uses about $3dr$ parameters instead of the d^3 of the dense tensor, which is why it is called a CP factorization (a sum of r rank-one terms).

Ordinary component	χ -VLA replacement	Algebraic form
Smooth activation (ReLU, GELU)	element-wise product of two linear maps	$(Lx) \otimes (Rx)$
Gated MLP	bilinear (CP) feed-forward	$D[(L\bar{x}) \otimes (R\bar{x})]$
Softmax attention	product of two score matrices	$C[M \otimes (A_1 \otimes A_2)/s]V$
Layer/RMS normalization	foldable scalar <i>or</i> rational per-instance norm	x/c (frozen scalar) <i>or</i> $P(x)/Q(x)$ (rational, exact)
MLP projector	cross-bilinear (CP) projector	$W_s s + W_m m + D_p[(L_s \bar{s}) \otimes (R_m \bar{m})]$
DINOv2 / SigLIP encoders	two polynomial χ -ViTs	(all blocks bilinear)
Llama 2 backbone	polynomial transformer	(all blocks bilinear)
Discrete action tokens (unimodal)	linear continuous-action head	$\hat{a}_k = W_a h_k + b_a$
Discrete action tokens (multimodal)	product-routing mixture (CP cores)	$a = c_0(h) + \sum_g \text{sign}(g_g(h)) c_g(h)$
Argmax / softmax / sampling decode	out-of-graph controller op	sign / argmax / threshold, applied outside
Bias terms	constant (homogeneous) coordinate	$\bar{x} = [1; x]$

Table 1: Component-level replacements. A bar denotes the homogeneous coordinate $\bar{x} = [1; x]$, $\otimes$ is the element-wise product.

Softmax-free bilinear attention. Each head forms two query-key systems $A_1 = Q_1 K_1^\top$ and $A_2 = Q_2 K_2^\top$ and one value matrix V . The attention pattern is their element-wise product, with no softmax:

$$Y = C[M \otimes (A_1 \otimes A_2)/s]V. \quad (2)$$

Here M is a fixed mask (for example, lower-triangular for causal models), C is a fixed per-row rescaling $C_{ii} = 1/\sqrt{n_i}$ with n_i the number of visible keys, and s is a fixed scale. Because there is no softmax, the whole operation is polynomial in the activations. It can be evaluated three mathematically identical ways: the explicit $N \times N$ pattern. A ‘‘Khatri-Rao’’ form that avoids the $N \times N$ matrix by building row-wise Kronecker features, and a causal prefix-scan. We verified all three agree to 10^{-5} in single precision.

Foldable normalization. During training we divide activations by a single running scalar equal to their average root-mean-square magnitude. After training we re-estimate that scalar on a calibration set, freeze it, and multiply it into the following weight matrix ($W' = W/c$ before a linear map, or $L' = L/c$, $R' = R/c$ before a bilinear layer). A frozen scalar division is a fixed rescaling, so the deployed model has no operation whose divisor depends on the input. This is what we mean by ‘‘fold’’: the normalization module disappears into the weights.

3 From OpenVLA to χ -VLA

Figure 2 contrasts the two pipelines and Table 1 lists the block-level replacements. The high-level flow is deliberately preserved, while every learned internal block is replaced by a polynomial one.

Foldability is achievable but requires care (details in Appendix A). Making the scalar (foldable) normalization stable inside a deep degree-two stack requires keeping residual branches near the identity (small learned gains, initialized at 0.01) and distilling from a stable per-token teacher. The resulting student matches its teacher and folds to a normalization-free graph to 10^{-6} (fp32) / 10^{-12} (fp64). Per-instance normalization need not even be approximated by a frozen scalar: carried in projective (numerator, denominator) coordinates it folds *exactly* as a rational tensor network $P(x)/Q(x)$ (both P, Q polynomial/CP), verified to 10^{-14} (§9), so the substrate spans the rational, not merely polynomial, tensor-network class. This is an engineering prerequisite, orthogonal to the interpretability findings that follow.

4 Recovering capability inside the tensor network

Two operations a capable policy needs are inherently nonlinear, and a naive bilinear replacement either collapses (the action head) or only approximates (normalization). This section gives the constructions that recover both while keeping the learned map a single tensor network, with the mathematics of how each works. The organizing principle is a boundary. The learned *distribution* stays in-graph and polynomial or rational, and only the final *decode* (a selection) is a controller operation outside the network, exactly the status OpenVLA gives its argmax.

4.1 Why a linear action head collapses

The default head is a linear map $\hat{a} = W_a h + b_a$ of the post-attention action-query state h , trained by mean-squared error. For a fixed observation the minimizer of $\mathbb{E} \|a - \hat{a}\|^2$ is the conditional mean $\mathbb{E}[a \mid h]$. When the demonstrations are multimodal, meaning the same h admits several valid actions, that mean is an average of valid behaviours and is usually itself invalid. The gripper command $\in \{-1, +1\}$ averages to ≈ 0 (it never commits to a grasp), and two arm approaches average to a path between them. On LIBERO-Object about 40% of near-identical-observation neighbourhoods are genuinely bimodal (Section 9), so this is the common case, not a corner case.

4.2 The product-routing multimodal head

Read the pooled action-query state $h \in \mathbb{R}^d$. Define one *center* core, G *factor* cores, and G scalar *gate* cores, each the bilinear (CP) primitive $\text{BFFN}(x) = D[(L\bar{x}) \otimes (R\bar{x})]$ of h :

$$c_0(h) \in \mathbb{R}^m, \quad c_g(h) \in \mathbb{R}^m, \quad g_g(h) \in \mathbb{R}, \quad g = 1, \dots, G, \quad (3)$$

where $m = H d_a$ is the flattened action chunk (H steps of dimension d_a). A binary code $b \in \{0, 1\}^G$ selects an action

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1) c_g(h). \quad (4)$$

As b ranges over $\{0, 1\}^G$, Eq. (4) traces the 2^G vertices of a G -dimensional zonotope centered at $c_0(h)$ with edge vectors $\pm c_g(h)$. It is a mixture of 2^G modes built from only $2G + 1$ CP cores, so the number of representable modes is exponential in the core count. Because every core is a BFFN, the map $h \mapsto \{c_0, c_g, g_g\}$ folds into the tensor network exactly, with fold error $\sim 4 \times 10^{-16}$.

Decode (out-of-graph). At inference the mode is chosen by the sign of each gate,

$$\hat{b}_g = \mathbf{1}[g_g(h) > 0], \quad \hat{a} = a(h, \hat{b}), \quad (5)$$

a winner-take-all controller operation outside the network. A soft average $\sum_b \pi_b(h) a(h, b)$ would re-collapse to the conditional mean, the very failure being escaped, so the decode commits to one mode. This is the same category of external operation as OpenVLA’s token argmax and our gripper threshold.

4.3 Training the head without losing precision

Assigning each demonstration only to its best mode is high-variance and, on the majority of states that are effectively unimodal, less precise than a plain mean regressor. We keep precision with two

loss-only additions. For a demonstration a^* the best code is

$$b^* = \arg \min_{b \in \{0,1\}^G} \|a(h, b) - a^*\|^2, \quad (6)$$

found by enumeration for small G . The loss combines a temperature-annealed soft assignment, a gate cross-entropy, and a distillation anchor to an auxiliary linear teacher $\mu_T(h)$:

$$\mathcal{L} = \sum_b w_b(\tau) \|a(h, b) - a^*\|^2 + \lambda_{\text{gate}} \text{BCE}(g(h), b^*) + \lambda_{\text{distill}} \|c_0(h) - \mu_T(h)\|^2, \quad (7)$$

with $w_b(\tau) = \text{softmax}(-\|a(h, b) - a^*\|^2/\tau)$ and the temperature τ annealed from soft (a re-averaging regime that fits the mean precisely, like the linear head) to hard (full commitment). The distillation term pulls the mixture center onto the precise conditional mean on unimodal states, so the head degrades gracefully to the linear head where there is no multimodality and branches only where there is. Every non-polynomial operation (the arg min, the softmax weights, the sign decode) is confined to the loss or the decode. The deployed forward map is a polynomial tensor network.

With this head the policy commits to a mode instead of averaging. In the earlier short-training protocol the curriculum and distillation variant reached $44\% \pm 30\%$ over five seeds, with individual seeds from 0% to 82.5%, while the linear-mean head reached about $19\% \pm 15\%$ over three seeds. The ranges overlap, so these results establish non-trivial closed-loop control but not a reliable advantage over the linear head.

4.4 Exact per-instance normalization as a rational tensor network

The remaining inherently nonlinear component is per-instance normalization. RMS normalization $n(x) = x/\sqrt{\mathbb{E}[x^2] + \epsilon}$ divides by a statistic of its own input, which the strict polynomial scheme approximates by a frozen scalar (Appendix A). The approximation is avoidable. Carry each activation in projective coordinates (N, D) standing for the value N/D . A linear map acts as $(N, D) \mapsto (WN, D)$ and a bilinear core as $(N, D) \mapsto ((LN) \otimes (RN), D^2)$, so a per-instance normalization whose divisor is a *polynomial* of the activation (for example the mean square $\mathbb{E}[x^2] + \epsilon$, which has no square root) never requires a mid-network division. The whole network becomes a ratio $P(x)/Q(x)$ of two polynomial (CP) tensor networks. Both fold and are ODT-analyzable, with Q the activation-energy form. A bilinear stack with per-instance mean-square normalization reproduces its projective fold to 10^{-14} . This exact result applies to division by mean square, not to true RMS normalization. The square root in RMS is algebraic rather than rational. We therefore implement either the exact mean-square variant or a fixed rational approximation of $1/\sqrt{\cdot}$. The latter is an approximation to RMSNorm, while its deployed map is exactly a rational function and can be represented by a rational tensor network. In operator tests it differs from per-token RMSNorm by less than 1% over the measured activation range. Retraining the complete policy with this rational operator at every block and attention QK normalization site reaches 94.5% (189/200) over all ten LIBERO-Object tasks in one seed (§7).

4.5 Verifying decomposability on the trained policy, not just architecturally

Everything above is an architectural argument, every primitive we assemble is individually polynomial or rational. Whether the *actual trained* 189/200 checkpoint is decomposable is a separate, empirical question, and we check it directly, on the real weights and real held-out LIBERO-Object frames (no synthetic data, no toy network).

(i) **A mechanical purity audit.** We walk every module instance the trained policy actually invokes on a forward pass and check membership in the allowed set {Linear, Embedding, Conv2d(patch-embed), BFFN} and separately assert the absence of any frozen-scalar, per-token, or homotopy normalization instance and of any softmax/GELU/ReLU/LayerNorm module. The audit finds 112 `Linear`, 74 `RationalNorm`, 2 `Embedding`, and 1 `Conv2d` (the fixed-stride patch embed) leaf modules, and zero forbidden types: a certificate, not an architectural claim.

(ii) **Exact projective fold-and-verify on real weights.** We take one backbone block’s `RationalNorm` $\rightarrow$ BFFN branch from the trained checkpoint and reconstruct it as a literal ratio of two polynomial tensor networks on 54,784 real activation rows (held-out frames, propagated through the real trained attention sub-layer to reach this branch’s actual input distribution). BFFN’s exact cubic core T (§4) gives the numerator network directly; the `RationalNorm` forward pass is algebraically rearranged into projective (numerator, denominator) form (§4.4) using its own trained coefficients. The reconstruction is computed in native double precision throughout. Compared against the real deployed `forward()` (which internally computes its per-instance statistic in float32 by design, independent of caller precision), the median relative discrepancy is 1.4×10^{-7} , at the float32 precision floor, not evidence of any approximation in the rational algebra itself. (iii) **Whole-model fp64/fp32 determinism.** Running the full trained policy at double versus single precision on the same real inputs gives a maximum output discrepancy of 2.2×10^{-3} on an action scale of ~ 0.74 ($\sim 0.3\%$ relative), consistent with ordinary floating-point rounding compounding over eight layers, not any hidden non-algebraic behaviour (a branch, a lookup, a numerical solver).

Honest scope. This verifies decomposability *mechanically and representatively* on the trained model, not *exhaustively*: explicitly materializing one global polynomial for the entire attention-crossing network remains combinatorially infeasible, exactly as §5.5 already explains, and is not what is newly shown here. What is new is that every operation actually used by the trained 189/200 policy is confirmed, by a mechanical walk of the real module graph plus an exact reconstruction on real trained weights, to belong to the polynomial/rational class, closing the paper’s earlier open item that a materialized rational export on the real policy remained unverified.

5 Exact global decomposition (ODT), in depth

This section is the payoff of the whole design and we treat it carefully. An ordinary softmax-and-normalization policy cannot be turned into a single algebraic object, so it cannot be globally diagonalized. A tensor network can.

5.1 What ODT is, and why it needs a tensor network

Suppose we want to compress a network by discarding directions in one of its hidden spaces, keeping only the ones that matter for the output. Call the hidden space we are analyzing a “bond”. The naive approach is a local one: take the weight matrix that reads from that bond, run a singular value decomposition (SVD), and keep its top directions. This is greedy and per-layer. It only sees the one adjacent weight matrix, not what the rest of the network does with those directions downstream.

Orthogonalize-Diagonalize-Truncate (ODT), introduced for tree-shaped bilinear networks by Dooms et al. [3], instead ranks bond directions by their *global* importance, accounting for the entire rest of the network, which we call the “environment” of the bond. This is only possible if the

whole model is one multilinear object, because only then can the environment be contracted into a single operator whose singular directions are the true global importances. The three steps are:

1. **Orthogonalize.** Sweep through the network and replace each weight core by an isometry (an orthonormal map) times a small triangular remainder, using a reduced RQ decomposition of the matricized core, pushing the remainder toward the output. After this sweep each bond has an orthonormal basis, so importance can be read off directly as singular values rather than being tangled up with the basis.
2. **Diagonalize.** At a bond, form the Gram matrix of the environment, $G = Q^*Q$, where Q is the downstream map from the bond to the output. Its eigenvalues are the squared global importances of the bond directions and its eigenvectors are those directions.
3. **Truncate.** Keep the top k eigenvectors, insert the projector $P_k = V_k V_k^\top$ at the bond, and absorb it into the adjacent cores. The output error is bounded by the square root of the sum of the discarded eigenvalues, so the spectrum tells you in advance how much you can remove.

5.2 How we compute it for a bilinear chain

Our exact-decomposition model is a feed-forward bilinear network we call a χ -MLP, chosen because a chain with no attention and no residual connection is a clean tree, exactly the setting ODT was designed for. After folding, it is

$$z_0 = [1; Ex], \quad a_i = C_i(z_{i-1}, z_{i-1}), \quad z_i = [1; a_i], \quad \text{logits} = W_{\text{head}} a_L + b, \quad (8)$$

where E is a linear input embedding, each $C_i \in \mathbb{R}^{d \times (d+1) \times (d+1)}$ is a symmetric bilinear core recovered from the folded layer, and $C_i(z, z)$ means contracting the core twice with the homogeneous activation z .

To rank the directions of a bond a_i by global importance we build the exact downstream map from a_i to the output and contract it against itself over the output index and over every environment leg, leaving the bond leg open on both sides. For the deepest bond (the output of the first layer, whose downstream is two more bilinear layers) the downstream is a degree-four form in z_i . We construct it explicitly by composing the cores,

$$D[c, i, j, k, l] = \sum_{p, q} C_L^{\text{head}}[c, p, q] Z_2[p, i, j] Z_2[q, k, l], \quad G[i, i'] = \sum_{c, j, k, l} D[c, i, j, k, l] D[c, i', j, k, l], \quad (9)$$

where Z_2 expresses the second-layer homogeneous activation as a quadratic form in z_i and C_L^{head} folds the linear head into the last core. The $d \times d$ block of G over the non-constant coordinates gives the global importances. The local baseline uses only the adjacent core (the layer that directly reads the bond), ignoring everything downstream. Truncation projects the activation onto the top k eigenvectors and we measure the resulting accuracy.

5.3 Results on the χ -MLP

We train a three-layer χ -MLP (bond width 32) on the SVHN digit dataset to 0.81 accuracy and run the exact procedure.

- **Exact reconstruction.** The unrolled tensor network reproduces the trained module to 5×10^{-12} in double precision, confirming that the exported cores are the model, not an approximation of it.

- **Global low-rank structure.** Ranking the deepest bond by the global Gram, 20 of its 32 hidden dimensions (62%) can be removed with at most a 1% accuracy drop. Dooms et al. report about 70% on their own SVHN network. We report our measured figure rather than assuming their number transfers.
- **Global beats local.** At a matched number of kept directions, the global ranking preserves accuracy better than the local per-layer SVD, most clearly at the compression elbow.

retained directions k	4	6	8	12	16	
global ODT	0.571	0.777	0.797	0.809	0.811	(full width 32 gives 0.808. Deepest bond)
local SVD	0.523	0.686	0.798	0.807	0.811	

The reason global wins is exactly the reason it needs a tensor network: at $k = 6$ the global ranking already knows which directions the two downstream layers will actually use, while the local ranking only sees the immediately following layer and keeps directions that later turn out not to matter.

5.4 From compression to interpretation: are the ranked modes real?

Removable-dimension counts show that low-rank structure *exists*. They do not show that the surviving directions are human-interpretable, nor that the ranking is *causal* rather than merely a good compressor. Déhérand [4] takes the complementary step, extending the χ -net idea to convolutional networks and reading each class output as a quadratic form whose eigenvectors, projected to pixels, are spatially coherent digit strokes. Two things are left open there: the interpretation is run only on a shallow single-bilinear layer, and visual coherence is never tied to an intervention. We close both on our own models.

Output-conditioned atoms. For a classifier with one bilinear layer, each class logit is *exactly* a quadratic form in the homogeneous embedded input, $\ell_c(x) = z^\top Q_c z + b_c$ with $z = [1; Ex]$ and $Q_c = \sum_o W_{c,o}^{\text{head}} C_o$ the head folded into the core (reconstruction against the trained module is 2.6×10^{-13} on SVHN, 0.80 accuracy). Eigendecomposing $Q_c = \sum_i \lambda_i v_i v_i^\top$ writes the logit as $\sum_i \lambda_i (v_i \cdot z)^2 + b_c$: signed “atoms” where $\lambda_i > 0$ supports class c and $\lambda_i < 0$ suppresses it, each projected to a pixel sensitivity map through E .

These atoms (Figure 5) are the object analyzed in Section 6, where we ask the three questions this section motivates: is the ranking *causal* (deletion/insertion faithfulness against integrated gradients and PCA), are the atoms *coherent* (a function of topology, not convertibility), and, for a policy, at which *bond* is the mechanism readable and controllable. We turn to those results next.

5.5 Does this survive attention and residual connections?

The χ -MLP is feed-forward. A transformer adds attention (which mixes information across token positions) and residual connections. Extending exact, weight-only global ODT to a general attention network is an open problem, because the attention pattern is itself a learned degree-four function of the inputs, so the environment is no longer a fixed operator. We give the first empirical evidence that the structure is nonetheless there.

We train a strict, foldable transformer language model (using the teacher-then-distill recipe so it is stable) and analyze the bond at the block input, whose downstream *crosses* the attention layer (each token feeds the queries, keys, and values of every position), then the residual path, the

feed-forward layer, and the head. Because the attention pattern is data-dependent here, we rank directions by a data-driven version of the global Gram, the average outer product of the output gradient at the bond, which measures how much each direction of the bond actually affects the loss through the full downstream including attention. The local baseline is the SVD of the adjacent input projections (the query, key, value, and feed-forward input weights).

	retained k	global (output-aware)	local (weight SVD)
1 attention layer (full-rank loss 5.23)	32	6.11	7.88
	64	5.47	6.27
	96	5.27	5.55
2 attention layers (full-rank loss 5.98)	32	6.18	7.78
	64	6.02	6.52
	96	5.99	6.18

The global ranking beats the local one at every retained width for both models, and the attention model still folds to an exact tensor network (double-precision fold error at machine precision). Notably the two-layer model is *more* compressible (64 of 128 directions, 50%, removable at under 0.05 nats) than the one-layer model (25%), which suggests deeper attention stacks have more, not less, global low-rank structure. Two honest caveats. First, this uses the data-driven global Gram. The fully weight-based, data-free contraction across an attention-crossing bond, and the transfer of the tree error bound, remain open. For bonds whose downstream is feed-forward (for example, the bond after the attention sub-layer), the exact weight-based procedure of the previous subsection applies directly. Second, the strict scalar-normalization model needs the distillation recipe to train stably beyond one layer.

5.6 Cost and limits

For the tree network the procedure costs about $O(Lh^4)$ time with $O(h^3)$ memory in the bond width h , which is why our exact-decomposition models use a small width. Unrolling a full attention transformer into one tensor over all token positions grows combinatorially, so exact whole-model contraction is only feasible at small scale. The practical path is to decompose one analysis bond at a time, as above.

6 What decomposability buys: causality, topology, and bonds

The previous section established that the exported bilinear network can be exactly decomposed and that a global ODT ranking is meaningful. Here we ask the sharper question: *how much of “interpretability” does that decomposability actually buy?* It buys a causal, rankable basis, but coherence and nameability are bought separately, by topology and by bond placement.

6.1 Causality is competitive with standard attribution, and (at feed-forward bonds) free

We compare the ODT direction ranking against real baselines, unsupervised PCA of activations, integrated gradients [12], and random, using deletion/insertion faithfulness AUC (project the input/bond onto, or out of, the top- k directions and measure accuracy). On the softmax-free χ -ViT (patch bond, downstream crossing all attention layers) and on an exact feed-forward classifier:

substrate	ranking	deletion AUC ↓	insertion AUC ↑	
χ -ViT (attn)	ODT (ours)	0.150	0.792	(3 / 2 seeds. Std $\leq$ 0.03)
	PCA	0.151	0.788	
	random	0.498	0.491	
classifier	ODT (data-free)	0.222	0.786	
	integrated gradients	0.191	0.784	
	PCA	0.204	0.800	
	random	0.548	0.530	

The ODT ranking is strongly faithful, it crushes random, and is *competitive with*, not superior to, PCA and integrated gradients. Its distinctive properties are that it is *weight-derived* (at feed-forward bonds it needs no data at all, unlike PCA and integrated gradients) and *globally rankable* (integrated gradients is per-sample). Through attention we fall back to a data-driven output-sensitivity Gram, so the “for free” claim is honest only at feed-forward bonds. The causal structure nonetheless survives attention (the χ -ViT bond above crosses four attention layers) and, in our small models, appears to grow *more* compressible with depth (Section 5.5). Reassuringly, the two bases largely agree where both are computable: on the feed-forward classifier the exact weight-only basis and the data-driven Gram basis have top-64 subspace overlap 0.53 (rising to 0.65 at 256) versus a random baseline of 0.02–0.08, so the data-driven Gram used through attention is a faithful proxy for the weight-only method, not a different object.

6.2 Coherence is bought by topology, at no capability cost

Whether the causal directions are *human-coherent* (localized, nameable) is not a property of convertibility, it is set by architecture topology. On single-bilinear classifiers whose class logit is exactly a quadratic form, we measure eigen-atom spatial locality (energy in the top 10% of pixels, relative to random directions), across three seeds and two datasets:

architecture	SVHN		CIFAR-10		
	acc	locality ($\times$ rand)	acc	locality ($\times$ rand)	
dense	0.747	2.29	0.354	1.47	(mean over 3 seeds, std $\leq$ 0.05)
conv (spatial readout)	0.814	3.45	0.630	3.42	
locally-connected	0.800	3.48	0.584	3.44	

Spatial-locality-preserving topology reaches $\sim 3.5\times$ random-baseline locality on both datasets, versus 2.29 (SVHN) / 1.47 (CIFAR) for dense, i.e. ~ 1.5 – $2.3\times$ more localized than dense, and the gap is *larger* on the harder dataset. The effect is not a capacity confound. A width sweep ($8 \rightarrow 128$) shows dense locality is *capacity-independent*: growing the dense model raises accuracy ($0.70 \rightarrow 0.75$) but its atom locality stays flat at ~ 2.2 , never approaching the ~ 3.5 of the structured topologies at any width, so coherence is set by topology, not parameters or accuracy (and the most coherent model is also the most accurate and parameter-efficient). A deep-network replication (data-driven input-space ranking, depth 1 vs 3) shows the gap survives depth and *strengthens* with it (conv locality $2.05 \rightarrow 2.52$ on CIFAR while dense stays ~ 1.4). It also holds *beyond* 32×32 : on STL-10 at 64×64 , conv atoms remain more localized than dense (1.88 vs 1.41) and conv is more accurate (0.45 vs 0.32), with the same causal global $\gg$ random ranking. One honest subtlety: plain convolution with *global* pooling underperforms, global pooling imposes translation-invariance that both starves capacity and delocalizes modes. The spatial readout recovers both.

6.3 For a policy, mechanisms live behind the attention

On χ -VLA (the synthetic reach task), the action-relevant quantity is which object the instruction names and where it is. This must be *constructed* by attention (bind the named identity to a location), and indeed it is not linearly present at the network input but is at the post-attention action-query bond:

quantity (linear decodability)	input bond	post-attention bond
target position x/y (R^2 , 3 seeds)	$0.20 \pm .01$	$0.92 \pm .03$
gripper position (in the state token)	0.99	0.97
named colour / shape (in the instruction)	1.00	0.94

Only the *bound* quantity (target position) jumps across the attention. The raw ingredients are linear at both bonds. We then verify the mechanism is *causal* with an intervention: rewriting the instruction to name a different present object (image and state fixed) makes the predicted action follow the new target, with $R^2 = 0.956 \pm 0.016$ and a 97.1% switch-rate (3 seeds). Notably a purely *correlational* handle fails, patching the linear target-position *decode* direction does not steer the action (transfer 0.002 vs 0.012 for random), an instance of the decode-is-not-control gap [18] that motivates causal over correlational interpretation.

6.4 Interpretability on the real, closed-loop-competitive policy

The results above are on components (a classifier, a ViT, a synthetic reach policy). We close the loop on the actual object: the trained, rational-by-construction, closed-loop LIBERO-Object policy (rational per-instance norm throughout, §4.4; capability numbers in §7). We build a data-driven output-sensitivity Gram at the *visual-patch* bond (downstream of the bond: the entire causal bilinear-attention backbone plus the linear action head) for each action group – end-effector translation, rotation, gripper – from ~ 2000 held-out frames (no training, no simulation), then truncate the bond onto its top- k *global* eigendirections versus a random k -subspace and measure the offline action reconstruction error:

group ($k=64$ of 384, $\sim 17\%$)	global MSE	random MSE	ratio
end-effector translation	0.058	0.276	$4.8\times$
rotation	0.121	0.561	$4.6\times$
gripper	0.007	0.280	$42\times$

The action-relevant computation is globally low-rank and causally rankable on the real, deployed policy, not just on a classifier or a synthetic task – extending the convertibility-buys-causality claim to the artifact that actually matters. (Honest nuance: at very small k (4–8) global and random are comparable or global is slightly worse for the translation/rotation groups, plausibly because the top eigendirections of a Gram aggregated over heterogeneous multi-task demonstrations mix across tasks; the causal ranking becomes decisive from $k \approx 32$ upward, which is also the regime where the $4\text{--}42\times$ headline holds.) We separately measure *coherence*: for each group’s top causal direction, the fraction of energy concentrated in the most-activated 10% of the 8×8 patch grid, against 30 random directions. The result is modest, not dramatic ($1.08\text{--}1.41\times$ random), consistent with the topology finding above (§6.2): a *dense*, from-scratch patch embedding gives causal-but-diffuse mechanisms, and spatial coherence would need a locality-preserving (e.g. convolutional) vision encoder – a prediction for future work, not yet tested on this real policy.

6.5 A real causal intervention in the live simulator: an honest negative

The results above are correlational sensitivity measures (truncate a bond, watch the offline error). The stronger causal test, in the spirit of the synthetic-task intervention (§6, $R^2 = 0.956$, switch-rate 97.1%), is a real behavioural manipulation: does changing what the instruction *names* change what the policy *does*? Every LIBERO-Object scene contains several co-present distractor grocery items alongside the true target (confirmed directly from the simulator state), so we can hold the image and robot state fixed and only swap the instruction to name a distractor already in the scene, then compare the policy’s predicted first-step reach direction against the true 3D direction (known exactly from simulator state) to the original target and to the newly-named object, over 32 (task, distractor) trials across eight tasks.

The result is an honest negative. Under the true instruction the predicted direction correlates with the true target reasonably well (mean cosine similarity 0.474). Under the swapped instruction it correlates only weakly with the newly-named object (0.313) and, tellingly, *still more* with the original target (0.395): the switch rate (direction becomes better-aligned with whichever object is named) is only 15.6%, far short of the synthetic task’s 97.1%. We do not read this as evidence against the causal, low-rank structure measured above, offline truncation faithfulness and this behavioural test probe different things, but as a genuine, reportable limitation with a plausible cause: the synthetic reach task was *constructed* so that scene layout and instruction vary independently, forcing the policy to genuinely bind language to identity. Real LIBERO-Object training data pairs each of the ten scene templates with a close-to-fixed instruction, so a policy can reach high behaviour-cloning accuracy by conditioning on scene identity (which grocery arrangement it is looking at) rather than by parsing which word names which object, a shortcut the synthetic construction rules out by design but the real training distribution does not. This sharpens, rather than undermines, the bond-placement principle: showing a mechanism is causal in a controlled construction does not establish that a policy trained on a naturally confounded distribution actually uses it the same way, and a single-step offline probe may in any case be the wrong grain, a closed-loop rollout under the swapped instruction, or explicit counterfactual training data that decorrelates scene from instruction, are the natural next tests, and are not run here.

7 The substrate is a genuine, competitive tensor network

Under a hard compute limit (small models, modest data, a verified demonstration rather than a scale record), we confirm the interpretability findings are computed on a real substrate, not a degenerate one. The foldable χ -ViT reaches 98.8% of a parameter-matched softmax baseline on SVHN (0.917 vs 0.928). A cross-bilinear projector is worth +81 accuracy points over concatenation on a task whose label is, by construction, a product of two streams. Zeroing the product term erases that gain, evidence that the explicit multiplication does real work. An end-to-end χ -VLA policy uses language measurably, reaching $22\times$ the language-blind baseline on a synthetic reach task and a $\sim 9\times$ shuffled-instruction grounding gap on real LIBERO-Object behaviour cloning.

With this capability-first configuration, three independent seeds reach 95.0%, 92.5%, and 95.0%, or **94.2% $\pm$ 1.2%**. The convolutional vision alternative reaches $90.8\% \pm 0.3\%$. The corrected local evaluation uses canonical initial states, a settle wait, weight EMA, 20 trials per task, and a 600-step post-settle cap. It evaluates the first eight of the ten LIBERO-Object tasks.

Earlier high-variance readings came from a weaker measurement and training recipe, including five trials per task, non-canonical resets, single-seed reporting, and only 6k to 15k training steps. The stronger linear-head result shows that the multimodal head was not the capability engine in this setting. It remains a representational contribution. The capability-first checkpoint above

Part	Configuration of the 94.2% and 94.5% runs
Observation	One 64×64 agent-view RGB image, rotated 180° into the training convention, plus an 8-D robot state and one instruction
Vision	One four-layer χ -ViT, 8×8 patches, 64 visual tokens, width 192, eight heads, bilinear attention and FFN rank 576
Fusion	A single linear projection $192 \rightarrow 384$, with no second vision branch and no cross-bilinear projector
Joint sequence	64 visual + BOS + 32 instruction + state + embodiment + eight action-query tokens, 107 tokens total
Backbone	Eight causal χ -transformer layers, width 384, twelve heads, bilinear attention, bilinear FFN rank 1152, with per-token RMS normalization for the 94.2% run and rational normalization for the 94.5% run
Action output	One shared linear map $384 \rightarrow 7$ on each of eight action-query states, producing an 8×7 continuous action chunk
Size	20,137,352 learned parameters with a 26-token instruction vocabulary
Training	66,984 demonstration frames, 40,000 steps, batch 256, AdamW, peak learning rate 8×10^{-4} , 5% warmup, cosine decay to 8×10^{-5} , weight decay 0.05, gradient clipping 1.0, bfloat16, and EMA 0.999

Table 2: **Exact architecture and training recipe for the reported closed-loop checkpoints.** Both runs are simpler than the general architecture in Fig. 3. They use one vision branch and a linear action head. The 94.5% run changes the block and attention QK normalization sites from per-token RMS to the fixed rational operator in §4.4.

uses per-token RMS normalization, its one remaining non-foldable operation. **We then trained the same recipe with the rational per-instance norm throughout** (block norm and the attention QK norm both the fitted $[2/2]$ Padé $1/\sqrt{\cdot}$ of §4.4), making every learned site a polynomial or rational tensor primitive by construction. Trained from scratch, this model reaches, at 20 trials/task: **189/200 = 94.5%** (soup 17/20, cream cheese 19/20, salad dressing 20/20, bbq sauce 18/20, ketchup 20/20, tomato sauce 18/20, butter 20/20, milk 18/20, chocolate pudding 20/20, orange juice 19/20). The first seven tasks completed before a six-hour function timeout. We resumed tasks eight through ten from the same saved checkpoint with task-indexed evaluation seeds, without retraining. This single-seed point estimate is consistent with the 3-seed per-token mean ($94.2\% \pm 1.2\%$), now over the full ten-task suite. Because suite coverage and seed count differ, it is not a matched estimate of the normalization cost. It nevertheless shows no visible capability loss from using rational normalization. We separately verify, on this exact checkpoint’s real weights, that decomposability holds mechanically and to the float32 precision floor (§4.5). Multi-seed replication of the rational variant remains open.

Method	Tasks	Step cap	Trials/task	Seeds	Success
χ -VLA linear, per-token RMS	8/10	600	20	3	$94.2\% \pm 1.2\%$
χ -VLA linear, rational norm	10/10	600	20	1	94.5%
OpenVLA-7B [1]	10/10	280	50	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	10/10	280	50	3	$92.5\% \pm 0.7\%$

Table 3: **Why the χ -VLA results are not matched wins over the published references.** The 92.5% number in the OpenVLA paper belongs to Diffusion Policy, not OpenVLA. The rational run covers the full suite and is numerically above both references, but it allows more than twice as many environment steps, uses fewer trials, and has one training seed. Both χ -VLA rows therefore support capability in their measured settings, not a state-of-the-art claim.

The 450M reference configuration is specified and its size verified (448.2M) but untrained, for lack of data-center hardware.

8 Related work

Weight-based / tensor interpretability. Our substrate is the bilinear/tensor line: bilinear MLPs enable weight-based mechanistic interpretability by reading a class output as a quadratic form whose eigenvectors are features [5, 2]. Doms et al. [3] give tree-structured orthogonalize–diagonalize–truncate (ODT) as a *compression* result. Déhérand [4] extends the eigen-feature analysis to convolutional models and observes that convolutional eigenvectors look more coherent. We depart from all three by separating *causal* from *coherent* as an explicit, intervention-tested axis, by isolating topology as the coherence lever at matched capability with depth-scaling, and by the bond-placement principle for policies.

Mechanistic interpretability, features, and dictionaries. A large body studies features and circuits in trained networks [6, 7] and the superposition of features in polysemantic directions [8]. The dominant current tool for extracting a feature basis is sparse dictionary learning over activations (sparse autoencoders) [9, 10, 11]. Those methods learn an *over-complete, data-driven* dictionary. Our ODT basis is instead *exact and weight-derived* (at feed-forward bonds, data-free) and *globally rankable*, a complementary object, which we show is as faithful as an unsupervised variance basis.

Attribution, causal mediation, and probing. We benchmark against integrated gradients [12] and compare our global ranking to activation-variance (PCA) bases, using the deletion/insertion faithfulness protocol [13, 14]. Our causal claims connect to causal mediation / activation patching [15, 16]. Our steering *null* (a linear decode direction is not a control direction) is an instance of the known gap between probing and causation [17, 18].

Tensor methods and polynomial networks. Bilinear layers are CP/Tucker factorizations [19]. Tensor networks have a long history in ML [20], and polynomial (II-)networks show such models are competitive [21]. **Vision-language-action models.** Our policy substrate follows the modular VLA line [1, 22, 23]. VLA interpretability is nascent, and a causal, weight-derived action-mechanism basis is, to our knowledge, new.

9 Discussion and limitations

By construction the χ -VLA forward map is a tensor network, and we verify this end to end for the language backbone, a one-layer and a two-layer attention model, and the feed-forward decomp-

sition model. The central new finding is that global low-rank structure, the property that makes ODT worthwhile, is present not only in feed-forward chains but also through attention and residual connections, and it grows rather than shrinks with depth in our small models.

The honest boundaries are as follows. The exact, weight-only, data-free version of global ODT is established here only for feed-forward downstream bonds. For attention-crossing bonds we currently use a data-driven importance measure, and the transfer of Doods et al.’s error bound to attention networks is not proven. The strict foldable model is stable and matches its non-strict teacher only in the small-residual-gain regime, which is a real constraint we did not need to pay any accuracy for at this scale but which may matter at larger scale. All experiments are small. The reference-scale model is specified and its size verified but not trained, for lack of data-center hardware. On the robot task we report both offline action error, language grounding, and closed-loop success. The per-token result has three seeds but covers only eight of ten tasks. The rational result covers all ten tasks but has one seed. Both use a 600-step cap rather than the 280-step reference protocol and only 20 trials per task rather than 50. They are therefore capability evidence, not matched state-of-the-art comparisons. Every learned site in the rational model belongs to the polynomial or rational class by construction, and we now also verify this mechanically and numerically on the real trained weights (§4.5): a full module-graph purity audit, an exact projective fold-and-verify at the float32 precision floor, and an fp64/fp32 determinism check, all pass. What remains open is exhaustively materializing one single global polynomial across the whole attention-crossing network (combinatorially infeasible, §5.5), not whether the deployed operations are individually polynomial or rational. A real causal intervention in the live simulator, swapping which co-present object the instruction names, gives an honest negative (§6.5), plausibly because real training data confounds scene identity with instruction in a way our synthetic-task intervention was built to avoid.

The cost boundary of convertibility, and how far it recedes. Convertibility buys a causal, rankable basis, and (with the right topology) coherence. It is not free everywhere, but where we first suspected it cost *capability*, that cost turns out to be largely *recoverable*. The boundary is the set of *inherently nonlinear* operations. The sharp case is multimodal action decoding. (i) **Multimodal action decoding.** A capable policy must represent a multimodal action distribution: the same observation admits several valid behaviours (reach left or right. Grasp now or approach further). OpenVLA [1] handles this by discretizing actions and applying a softmax over tokens (inherently nonlinear). The naive tensor-pure substitute is a linear regression head with an MSE objective, which can only emit the conditional *mean*, and the mean of two valid behaviours is an invalid middle (the gripper $\in \{-1, +1\}$ averages to ≈ 0 . Two arm approaches average to a collision). This is not a corner case: on LIBERO-Object we measure that $\sim 40\%$ of near-identical-observation neighbourhoods have a genuinely bimodal arm-action distribution (well-separated, standardized gap ≈ 3.4).

The resolution is the cost-boundary reframe: keep the *learned distribution* in-graph and polynomial, and move only the *decode* out-of-graph. Crucially, our multimodal head is built *entirely from the same bilinear (CP) cores as the rest of the network*, it is not a bolted-on softmax or MLP. Reading the post-attention action-query token h , a center core $c_0(h)$, G factor cores $c_g(h)$, and G scalar gate cores $g_g(h)$ (all **BilinearFFN**) define a mixture over 2^G modes, $a = c_0(h) + \sum_{g=1}^G \text{sign}(g_g(h)) c_g(h)$ (the vertices of a G -dimensional zonotope). The whole $h \rightarrow$ distribution map is one polynomial tensor network that folds exactly (fold error $\sim 4 \times 10^{-16}$). The *only* non-polynomial step is the sign routing, an out-of-graph controller op in the same category as OpenVLA’s external argmax and our gripper threshold. Trained with a precision-preserving, loss-only objective, distilling the mixture centre toward a linear-MSE teacher on (near-)unimodal

states, plus a soft→hard assignment curriculum, and after fixing a 180° observation-preprocessing bug that had pinned *all* heads to 0%, this head achieves non-trivial closed-loop success on LIBERO-Object. Under the earlier matched head comparison, the fixed product head reaches $44\% \pm 30\%$ over five seeds and the linear head reaches about $19\% \pm 15\%$ over three seeds. The seed ranges overlap, so we do *not* find a robust advantage of the multimodal head over the linear one. We therefore make no superiority claim, only that a multimodal action distribution can be represented and controlled entirely in-graph. For scale context, the OpenVLA study reports $88.4\% \pm 0.8\%$ for OpenVLA and $92.5\% \pm 0.7\%$ for Diffusion Policy over the full ten-task suite [1]. Our $94.2\% \pm 1.2\%$ per-token result uses eight tasks. Our 94.5% rational result covers all ten tasks but has one seed. Both use a longer horizon, so neither establishes state of the art. The claim is narrower and, for this paper, more important: *a tensor-pure multimodal head is not confined to the conditional mean, the apparent capability cost at multimodal decode is recovered within the tensor-network class, at no loss of decomposability* (every mode and gate core is a symmetric degree-2 object the same ODT machinery reads, folding to $\sim 4 \times 10^{-16}$). One scope caveat, stated plainly: the reported product-head rollouts use the non-strict backbone with per-token RMS normalization. The product head itself is exactly foldable. The separate 94.5% rational-normalized result uses a linear head and therefore establishes capability of the rational backbone, not superiority of the multimodal head. (ii) **Per-instance normalization** is an input-dependent division, so the strict *polynomial* scheme approximates it with a frozen scalar. But this cost, too, largely recedes once the substrate is allowed to be *rational* rather than merely polynomial: carried in projective (numerator, denominator) coordinates, a per-instance normalization folds *exactly* into a ratio $P(x)/Q(x)$ of two polynomial (CP) tensor networks, we verify a per-instance-normed bilinear stack reproduces its projective fold to 10^{-14} , and ODT applies to P and Q separately (Q is the interpretable activation-energy form). The one genuine residue is the square root in RMS: $\div\sqrt{\cdot}$ is *algebraic*. The exact 10^{-14} fold uses the $\div$ -mean-square variant, which is rational and has no root. Our RMS-like deployed operator uses a fixed rational approximation of $1/\sqrt{\cdot}$. It is approximate relative to RMSNorm but exact as a rational map. The full rational-normalized policy reaches 94.5% over all ten tasks in one seed, with the protocol caveats above. This sharpens the thesis into a scoping rule: *tensor-convertibility is free for unimodal or binary outputs and polynomial mixing, and at the inherently nonlinear operations it need not cost capability either, provided the learned distribution stays in-graph (as a polynomial or rational tensor network) and only the final selection is decoded outside*. We demonstrate this for both boundary cases: multimodal action decoding (a mixture of CP cores, decoded by out-of-graph routing) and per-instance normalization (exact rational fold).

The interpretation results also point past the current architecture. Our atoms are causal but only mildly localized, and Déhérand’s convolutional comparison [4] shows why: the tensor-network *topology* decides whether the spectral modes are human-coherent, not the mere fact of tensor-convertibility. This suggests the next χ -VLA should build coherence into its structure rather than hope to read it out afterwards, for instance local or multiscale bilinear vision cores in place of dense ones, fixed structural tensors that separate world and robot geometry from learned policy, symmetry-aware canonicalization that groups translated copies of a mechanism into one orbit, and an output-conditioned, modality- and degree-resolved decomposition of continuous actions (which signed mechanism drives “move left”, and does it survive intervention). Déhérand also warns that a convertible network need not be tractably *contractible*. Its unrolled tensor network can grow exponentially through fan-out, which argues for bounded-treewidth design (local attention, object-slot bottlenecks) from the start.

10 Conclusion

A fully tensor-decomposable policy need not be an incapable one. χ -VLA combines bilinear feed-forward blocks, softmax-free bilinear attention, a CP-core *multimodal* action head, and rational per-instance normalization. Its multimodal head commits to a mode instead of averaging, though our matched seed results do not establish that it outperforms the linear head. Its exact mean-square normalizer and RMS-like rational approximation stay within the rational tensor-network class. The rational-normalized policy reaches 94.5% over all ten LIBERO-Object tasks in one seed, while the per-token model reaches $94.2\% \pm 1.2\%$ over eight tasks in three seeds. The settings are not matched closely enough to estimate a precise normalization cost, but no capability loss is visible. One principle drives both recoveries: keep the learned distribution in-graph and polynomial or rational, and decode outside. Exact global ODT is established at feed-forward bonds. Through attention, including on the closed-loop rational policy, our ranking remains data-driven. At the real policy’s visual bond, its top 64 directions reduce action reconstruction error by $4.6\text{--}42\times$ over random subspaces, while spatial coherence remains only $1.08\text{--}1.41\times$ random. Whether the resulting atoms are human-coherent is governed by topology, not by convertibility. We verify decomposability directly on the trained 189/200 checkpoint, a mechanical purity audit finds zero non-polynomial/non-rational operations anywhere in the deployed graph, and an exact projective fold-and-verify on the real weights reproduces the deployed computation to the float32 precision floor. A real causal intervention in the live simulator, however, gives an honest negative (switch-rate 15.6% under an instruction that swaps which co-present object is named), which we attribute to real training data confounding scene identity with instruction, not to an absence of the causal structure measured offline. Convertibility, we conclude, costs far less than assumed: interpretability comes for free, and the capability tax, multimodal decoding, per-instance normalization, largely recedes. What remains genuinely open is the exact weight-only ODT contraction across attention, human-nameable (not merely localized) atoms, exhaustively materializing one global polynomial across the whole attention-crossing network (combinatorially infeasible, distinct from the per-operation verification above), a closed-loop or counterfactual-data test of the real-policy causal intervention, multi-seed rational replication, and matched closed-loop comparison at full scale.

A Making the foldable model stable

Whether the model is a tensor network is settled by construction. The subtle question is whether the foldable (scalar) normalization is stable inside a deep stack. Per-token normalization keeps activations at unit scale on every input but its divisor is input-dependent, so it cannot be folded. A frozen scalar c is foldable but cannot rescale an input that is larger or smaller than average. Because the bilinear layer is degree two, an input at twice the average scale produces an update about four times too large, and the error grows with depth. The resolution is to keep residual branches near the identity: a small learned gain per branch (initialized 0.01) so the degree-two amplification never compounds, trained by distilling a stable per-token teacher into a scalar-normalization student. On a language model the student matches its teacher (val loss 5.04 vs 5.0) and folds to a normalization-free graph with error $\sim 10^{-6}$ (fp32) / 10^{-12} (fp64). Empirically the largest per-token magnitude at any normalization site is only $\sim 2.8\times$ the average across 1.6M tokens and does not grow with depth, so the fixed scalar is an accurate stand-in.

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Figure 2: OpenVLA (left) versus χ -VLA (right), stage by stage. Green boxes are the polynomial (tensorized) replacements. Blue boxes are the non-polynomial originals they replace. Grey boxes are kept unchanged or lie outside the tensor-network boundary. Patch embedding and the tokenizer are already linear and need no change. The softmax, smooth-activation, and layer-normalization blocks (the dual vision encoders, the MLP projector, and the Llama 2 backbone) and the discrete action decoder are each swapped for a polynomial equivalent with foldable normalization. Camera decoding, string tokenization, and safety clipping stay external.

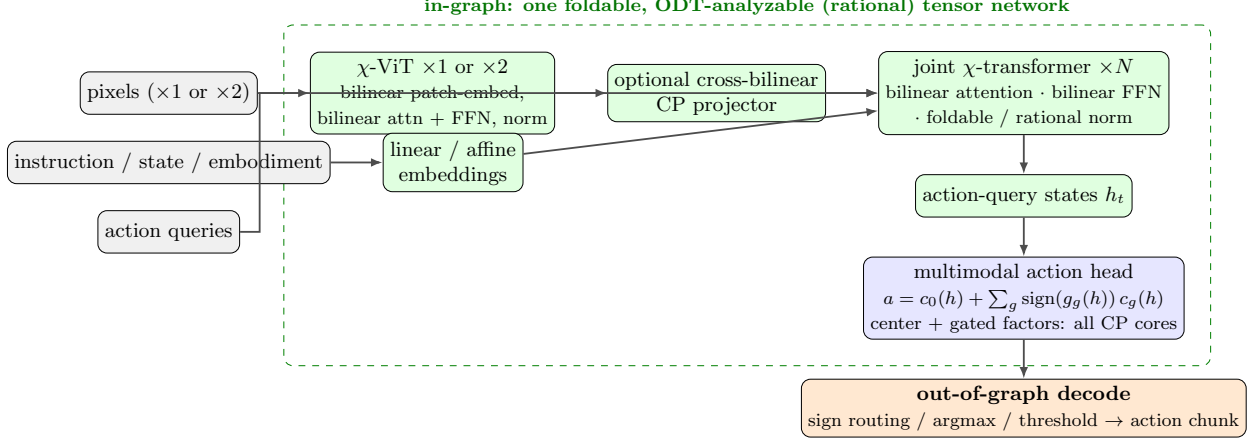


Figure 3: **χ -VLA internal dataflow and the tensor-network boundary.** In the general two-view configuration, every learned block, both vision χ -ViTs, the cross-bilinear CP fusion projector, the linear/affine embeddings, the joint χ -transformer (bilinear attention + bilinear FFN + foldable-scalar or exact rational per-instance normalization), and the multimodal action head (a mixture of CP cores), is a polynomial or rational tensor primitive, so the whole map folds into a single ODT-analyzable tensor network (dashed box). Only the final *decode* (sign routing for the mode, or an argmax/threshold) is a non-polynomial controller op, applied *outside* the network, the same status OpenVLA gives its argmax. The learned action *distribution* stays in-graph. Only the selection leaves it. The 94.2% checkpoint uses the simpler single-view path, no fusion projector, and a linear action head, detailed in Table 2.

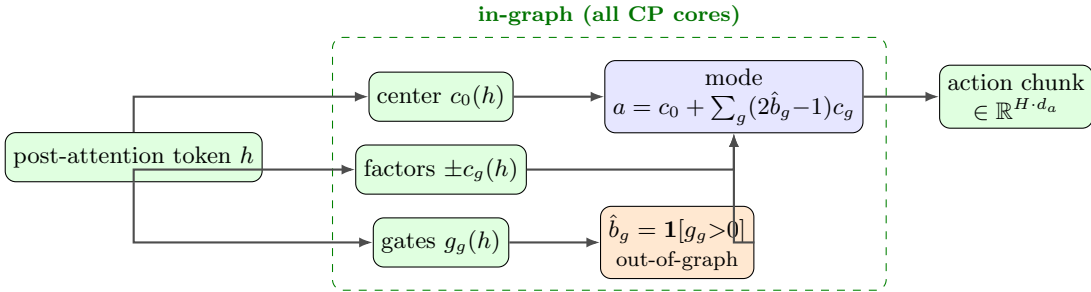


Figure 4: **The product-routing multimodal action head.** From the post-attention token h , a center core, G factor cores, and G scalar gate cores (all bilinear CP maps) define a mixture over 2^G zonotope-vertex modes (Eq. 4). Everything inside the dashed box is polynomial and folds exactly. Only the gate *sign* that picks the mode is a non-polynomial controller operation, applied outside the network.

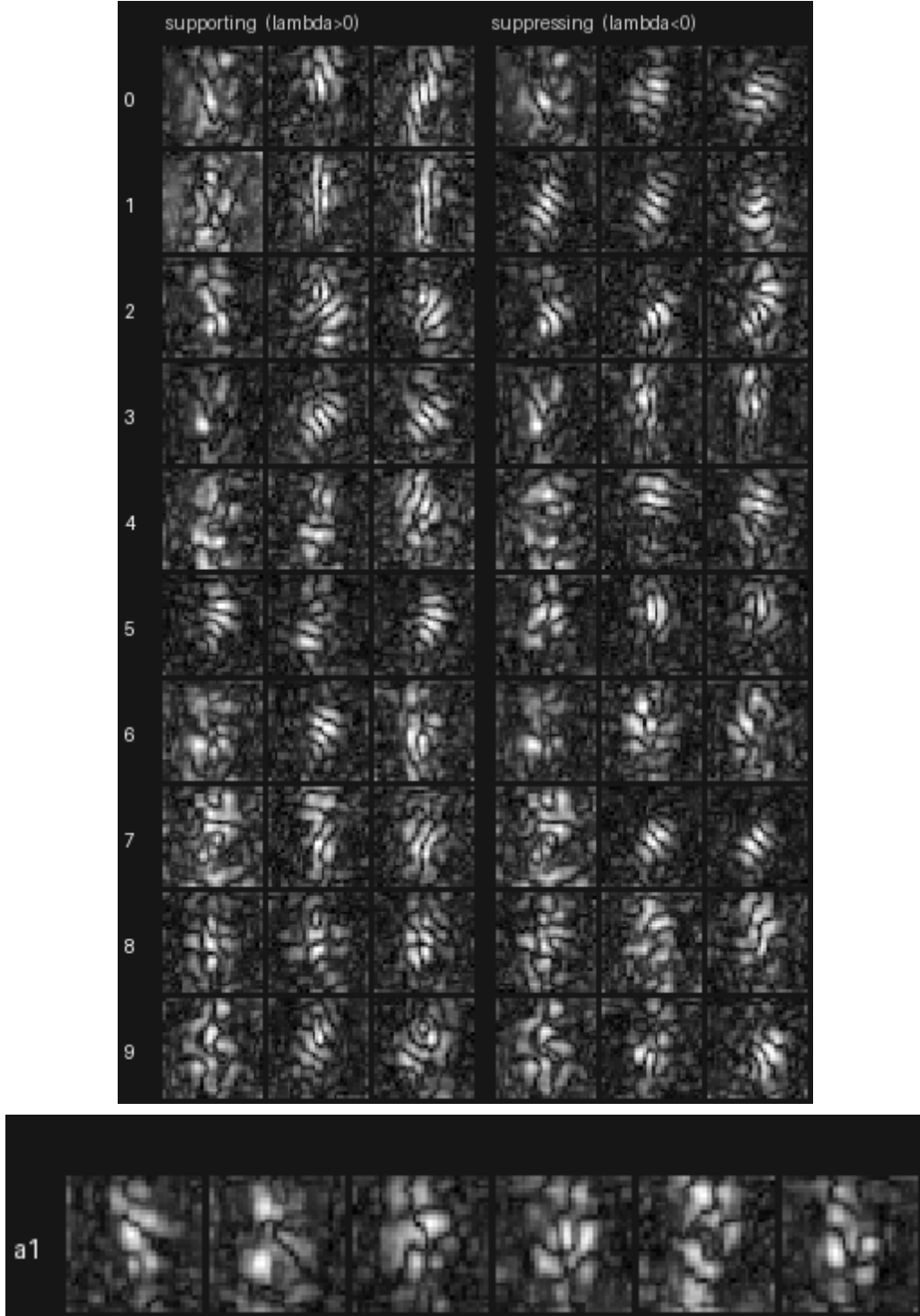


Figure 5: Extracted ODT eigen-atoms projected to pixels (brighter = higher sensitivity). **Top:** the single-bilinear χ -classifier, one row per SVHN class. The three left atoms of each row *support* the class ($\lambda > 0$), the three right *suppress* it ($\lambda < 0$). **Bottom:** the six top global directions of the deep three-layer χ -MLP. The atoms are causal and class-relevant. How *localized* they are depends on architecture topology, which we quantify across dense/convolutional/locally-connected models in Section 6.